

Data Cleaning Steps

End-to-End Retail Performance and Behavioral Analytics

Tool Used: Python (Pandas)

1. Project Objective

The objective of data cleaning was to prepare the raw retail datasets for analysis by removing inconsistencies, correcting data types, handling missing values, and creating clean datasets suitable for SQL analysis and Power BI visualization.

2. Datasets Used

The project contains five raw datasets:

- sales_data.csv
 - customers.csv
 - products.csv
 - stores.csv
 - returns.csv
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3. Data Import

Imported all datasets into Python using the Pandas library.

```
import pandas as pd

sales = pd.read_csv("sales_data.csv")
customers = pd.read_csv("customers.csv")
products = pd.read_csv("products.csv")
stores = pd.read_csv("stores.csv")
returns = pd.read_csv("returns.csv")
```

4. Initial Data Exploration

Performed an initial assessment of each dataset by:

- Viewing the first few records.
- Checking the number of rows and columns.
- Inspecting data types.
- Identifying missing values.
- Detecting duplicate records.

Python functions used:

- `.head()`
 - `.info()`
 - `.describe()`
 - `.isnull().sum()`
 - `.duplicated().sum()`
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5. Handling Missing Values

Missing values were identified in all datasets where applicable.

Cleaning activities included:

- Removing unnecessary null records.
 - Retaining valid null values where business rules allowed (e.g., Online sales without a Store ID).
 - Verifying dataset completeness after cleaning.
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6. Removing Duplicate Records

Duplicate rows were detected and removed using:

```
drop_duplicates()
```

This ensured each record represented a unique transaction or entity.

7. Data Type Conversion

Converted columns to appropriate data types.

Examples include:

- Order Date → Date
- Signup Date → Date
- Return Date → Date
- Quantity → Integer
- Unit Price → Decimal
- Total Amount → Decimal

Proper data types improved analytical accuracy and dashboard performance.

8. Standardizing Data

Performed formatting and consistency improvements:

- Removed extra spaces from text columns.
 - Standardized column names.
 - Ensured consistent capitalization.
 - Corrected formatting inconsistencies.
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9. Feature Engineering

Additional columns were created to support business analysis.

Profit

Calculated using:

`Profit = Total Amount - (Cost Price × Quantity)`

Order Year

Extracted from Order Date.

Order Month

Extracted from Order Date.

Age Group

Customers were grouped into categories such as:

- 18–24
- 25–34
- 35–44
- 45–54
- 55+

These derived columns simplified reporting and dashboard development.

10. Data Validation

After cleaning, the datasets were validated by:

- Rechecking missing values.
- Confirming data types.
- Ensuring row counts remained consistent.
- Verifying calculated columns.
- Confirming key relationships between datasets.

11. Exporting Cleaned Data

The cleaned datasets were exported as CSV files for use in SQL Server and Power BI.

Generated files:

- cleaned_sales.csv
- cleaned_customers.csv
- cleaned_products.csv
- cleaned_stores.csv
- cleaned_returns.csv

12. Outcome

The cleaned datasets were:

- Free from duplicate records.
- Properly formatted.
- Consistent across all tables.
- Ready for SQL analysis.
- Optimized for Power BI dashboards.

These cleaned datasets formed the foundation for business analysis, KPI calculation, and interactive dashboard development.
